

BIS RAG Knowledge Base — Source Document (v2, Extended)

Product: Self-Ballasted LED Lamps for General Lighting Services | Standard: IS 16102 | Scheme: CRS (Scheme-II)

What's new in v2: Legal order text (2012 gazette establishing CRS), the 2026 gazette notification establishing IS 16102/10322/16614 2026 editions with exact concurrent-running dates, Standard Mark specification & labelling rules, full registration/renewal/inclusion process, ETD 23 technical committee composition, classification metadata, direct BIS contact details, 53-question official FAQ (product-coverage edge cases), and the MeitY Applicant Testing Handbook (with an explicit scope-limitation note for LED lamps).

Compiled: 02 September 2026 | Prototype: SIH26 BIS AI Assistant (Team Byte Busters)

SECTION A — Standard Classification (Addendum)

[CHUNK TYPE: structured-metadata] [RAG-TAG: classification, ministry, ics_code]

Field	Value
Group	Electrical Appliances and Accessories
Sub-Group	Lamps, Luminaire and Accessories
Sub-Sub-Group	Luminaires & Lighting
Relevant Ministry	Ministry of Power
ICS Code	ICS 29.140.99
Identical/Equivalent International Standard	IEC 62560:2015
Certification (field)	N/A (certification is via CRS registration, not a direct ISI field)
Sustainable Development Goals tag	N/A (not tagged on this standard)

SECTION B — Legal Foundation: Electronics & IT Goods (Compulsory Registration) Order, 2012

[CHUNK TYPE: regulation-text] [RAG-TAG: legal_basis, crs_order_2012, prohibition_clause]

Source: The Gazette of India, Extraordinary, Part II–Section 3(ii), No. 1975, dated 3 October 2012. Issued by the Department of Electronics and Information Technology, Ministry of Communications and Information Technology, under Section 10(1)(p) of the Bureau of Indian Standards Act, 1986, read with Rule 13(fa) of the BIS Rules, 1987. Order signed by Dr. Ajay Kumar, Joint Secretary. Published in Hindi and English.

Key provisions (paraphrased from the Order):

- **Short title & commencement (Clause 1):** The Order is titled the 'Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order, 2012', and came into force six months after its gazette publication.
- **Definitions (Clause 2):** Defines 'Appropriate Authority' (an officer not below Director/Scientist-F of DeitY or its offices, authorised by the Secretary), 'Goods' (items listed in Schedule column 2), 'Series of products' (same family varying in size/rating/construction), and 'Specified Standard' (the IS listed in Schedule column 3).
- **Core prohibition (Clause 3):** No person may manufacture, store for sale, import, sell, or distribute any listed good unless it conforms to the specified Indian Standard AND bears the words 'Self Declaration — Conforming to IS (relevant standard)' after obtaining registration from BIS. This prohibition does not apply to goods manufactured for export.
- **Non-conforming goods (Clause 3(2)):** Substandard or defective goods not conforming to the specified standard must be deformed beyond use by the manufacturer and disposed of as scrap — they cannot simply be pulled from sale and resold later.
- **Power to call for information (Clause 4):** The Appropriate Authority (or a person it authorises) may demand records/samples relating to manufacture, storage, sale or distribution. With specific senior-level authorization, it may also inspect books/goods, and enter, search, and seize goods believed non-conforming, applying Code of Criminal Procedure, 1973 search-and-seizure provisions.
- **Sampling and testing (Clause 5):** Products with different sizes/ratings/varieties may be grouped for 'series approval' based on representative model testing (decision taken by DeitY or its authorised agencies). Samples must be drawn at least once every two years per registered product/series, chosen randomly from manufacturing unit or market, sealed and labelled with registration number, sample date, source, and a three-letter code identifying the sampler. Where sampled from the manufacturing unit, an additional counter-sample from the same batch is sealed and left with the registered user for later dispute use. Sealed samples go to a BIS-recognised lab for testing against all requirements of the specified standard; conforming test reports are retained by the Appropriate Authority, non-conforming ones are forwarded to BIS for action, and processing-status information must be published on the Authority's website.
- **Power to issue directions & manufacturer obligations (Clauses 6–8):** The Appropriate Authority may issue directions to manufacturers as necessary to carry out the Order; every manufacturer must comply, and must not refuse to furnish lawfully demanded information under Clause 4, nor destroy, mutilate, or deface related books/documents in its possession or control.

Original Schedule (2012) — 15 product categories, all with 03 July 2013 implementation: Electronic Games (Video) — IS 616:2010; Laptop/Notebook/Tablet — IS 13252:2010; 32"+ Plasma/LCD/LED TVs — IS 616:2010; Optical Disc Players with 200W+ built-in amplifier — IS 616:2010; Microwave Ovens — IS 302-2-25:1994; Visual Display Units/Monitors 32"+ — IS 13252:2010; Printers/Plotters — IS 13252:2003; Scanners — IS 13252:2010; Wireless Keyboards — IS 13252:2010; Telephone Answering Machines — IS 13252:2010; Amplifiers 2000W+ — IS 616:2010; Electronic Musical Systems 200W+ — IS 616:2010; Electronic Clocks with Mains Power — IS 302-2-26:1994; Set Top Box — IS 13252:2010; Automatic Data Processing Machines — IS 13252:2010. LED Lamps were added later via the 2014 amendment (see Section 3), effective 13 September 2015.

SECTION C — 2026 Standard Establishment & Concurrent-Running Notification

[CHUNK TYPE: regulatory-notice] [RAG-TAG: freshness, concurrent_running, withdrawal_date] [HIGH VALUE — this directly powers the status-badge feature]

Source: The Gazette of India, Extraordinary, Part III–Section 4, No. 95, dated 10 February 2026. Ref: HQ-PUB013/1/2020-PUB-BIS (1420). Issued by BIS, Department of Consumer Affairs, under Rule 15(1) of the BIS Rules, 2018. Published bilingually (Hindi/English), CG-DL-E-13022026-270101.

What it says: BIS notifies that the standards listed in column 2 of the schedule were established on the date in column 3. Where an older standard is listed in column 4, that older standard remains concurrently valid until the withdrawal date in column 5 — i.e. there is a transition window where BOTH editions are technically in force.

New Standard Established	Est. Date	Old Standard (concurrently valid until withdrawal)	Withdrawal Date
IS 10322 (Part 1):2026 — Luminaires General Req. & Tests (IEC 60598-1:2024, MOD) (2nd Rev.)	02 Feb 2026	IS 10322 (Part 1):2014 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 1):2026 — Fixed General Purpose Luminaires (2nd Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 1):2012 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 2):2026 — Recessed & Recessed Air-Handling Luminaires (2nd Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 2):2012 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 3):2026 — Road/Street Lighting (2nd Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 3):2012 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 4):2026 — Portable General Purpose Luminaires (1st Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 4):1987	02 Aug 2026
IS 10322 (Part 5/Sec 5):2026 — Floodlights (2nd Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 5):2013 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 6):2026 — Handlamps (1st Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 6):2013	02 Aug 2026
IS 10322 (Part 5/Sec 7):2026 — Lighting Chains (2nd Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 7):2017 (1st Revision)	02 Aug 2026
IS 10322 (Part 5/Sec 8):2026 — Emergency Lighting (1st Rev.)	02 Feb 2026	IS 10322 (Part 5/Sec 8):2013	02 Aug 2026
IS 10322 (Part 5/Sec 18):2026 — Swimming Pool Luminaires (NEW, no prior edition)	02 Feb 2026	N/A	N/A
IS 16102 (Part 1):2026 — Safety Requirements (1st Revision)	02 Feb 2026	IS 16102 (Part 1):2012	02 Aug 2026
IS 16102 (Part 2):2026 — Performance Requirements (2nd Revision)	02 Feb 2026	IS 16102 (Part 2):2017 (1st Revision)	02 Aug 2026
IS 16614 (Part 1):2026 — Double-Capped LED Linear Lamps, Safety (1st Rev.)	02 Feb 2026	IS 16614 (Part 1):2018	02 Aug 2026

Practical rule for the assistant to apply: As of today (2 September 2026), IS 16102 (Part 1 & 2):2012/2017 have already passed their 02 August 2026 withdrawal date and are now formally withdrawn — only the 2026 editions are valid. However, recall the separate BIS circular (Section 1.1 in the base v1 document) extending the *implementation* deadline for manufacturers to 02 February 2027. These are two different things: the OLD standard's legal validity ended 02 Aug 2026, but manufacturers get extra time (until 02 Feb 2027) to fully transition their registrations to the new standard. The assistant should explain both dates if asked 'is my 2012-standard product still compliant?' — legally no (standard withdrawn), but a grace/implementation period is running.

Signed by: Chitra Gupta, Scientist G & DDG (Hallmarking and Training)

SECTION D — Standard Mark & Labelling Requirements

[CHUNK TYPE: regulation-explanation] [RAG-TAG: standard_mark, labelling, r_number_format]

Source: BIS Central Marks Department-3, Ref: CMD 3/8:1/6975, dated 03.12.2015, signed by C.K. Maheshwari, DDG (Certification). Standard Mark notified via Gazette S.O. 3240(E) dated 01 December 2015. This circular supersedes the earlier order BIS/DGO/(405)/2014 dated 31 July 2014.

Key rules:

- The Standard Mark monogram is a fixed pictorial design (an 'S' formed of interlocking blue and red arrows) drawn in the exact style set out in Annexure I — photographic enlargement/reduction is permitted, but the design itself is not.
- It can be shown in single colour (no colour restriction), or in the official multi-colour scheme of Red, Blue and Black: Oriental Blue and Monopol Red for printing purposes (per IS 1222:1992), or French Blue (No.166) and Red (No.537) for sign-board purposes (per IS 5:1994).
- Every registered product must display either the Standard Mark OR the words 'Self Declaration — Conforming to IS...' — both paired with the Registration Number in the format R-XXXXXXXX (8 digits).
- Display: on the article and/or packaging, in a manner easily visible, legible, indelible, and non-removable; durability must follow the relevant Indian Standard's marking-durability provisions where applicable. Minimum text size for IS number, registration number, and words: Arial font size 6.
- Manufacturers may freely choose either the graphical Standard Mark or the plain text declaration — both are equally valid compliance routes; there is no requirement to use one over the other.

Registration number format confirmed: R-XXXXXXXX (8 digits after 'R-'), matching the real R-numbers already collected in Section 5 of the base v1 document (e.g. R-41014095).

SECTION E — Registration, Renewal & Inclusion Process

[CHUNK TYPE: process-explanation] [RAG-TAG: certification_process, renewal, inclusion]

E.1 — New Registration (Steps involved)

1. Apply online for grant of Registration.
2. Submit original test reports from BIS-recognized CRS labs with the application.
3. Foreign applicants without an Indian liaison/branch office must appoint an Authorized Indian Representative (AIR) via the prescribed nomination form.
4. AIRs of foreign applicants submit an affidavit-cum-undertaking (Format A/B); domestic applicants submit Format C.
5. Deposit the prescribed CRS fee.
6. Fill Form VI online, per the documented checklist.
7. Submit hard copies of the application within 15 days of online submission — non-receipt within 15 days closes the application automatically.
8. BIS scrutinizes the application; queries are raised and must be answered online without delay.
9. Once queries are resolved and the affidavit is submitted, Registration is granted and the grant letter is uploaded to the applicant's portal.

Timeline: BIS follows First-Come-First-Serve processing from hard-copy receipt date. Normal time for grant of Registration: **20 working days**.

E.2 — Renewal (every 2 years)

Registration is initially valid for 2 years and must be renewed thereafter (renewal application plus fee must be submitted at least one month in advance of expiry). Process: click 'Request renewal' on the dashboard → fill production report details → declare quantity marked with Self Declaration/Standard Mark → upload distributor list PDF → upload Form IX → pay fee → (foreign applicants) confirm AIR compliance as above → submit hard copies within 15 days. Per the FAQ, repeat testing is not specifically required for renewal, though test samples may be picked up during surveillance, and retesting may be required per para 3 of the Order dated 13 Nov 2014 if a test report per the latest upgraded standard is unavailable.

E.3 — Inclusion of New Models (under an existing Registration)

Registered manufacturers can add new models of the same product/brand under their existing R-number. Process: test the new model at a BIS-recognized lab per MeitY series guidelines → test report must be under 90 days old at submission → sign in → 'Request model inclusion' → upload test report, undertaking, fee details → submit online, then hard copies within 15 days. **Normal time for grant of inclusion: 10 working days.**

Visiting hours for AIR/domestic representatives at the Registration Department: 3 PM – 5 PM, Monday to Friday (photo ID and authorization letter required).

SECTION F — Technical Committee: ETD 23 (Lamps and Related Equipments)

[CHUNK TYPE: structured-metadata] [RAG-TAG: technical_committee, etd23]

Field	Value
Department	Electrotechnical Department (ETD)
HOD	Asit Kumar Maharana
Member Secretary	Jainendra Kumar
Scope	To prepare standards for all types of electric lamps, caps and their auxiliaries
Published Standards	67
Standards Under Development	7
Total Members	31 (Large Scale Industry: 9, R&D; Organization: 5, Central Ministry/Dept: 5, Consumer Group: 3, Regulatory Body: 3, Academic Institution: 2, In Personal Capacity: 2, Industry Association: 1, State Government: 1)
Notable member organizations	Bajaj Electricals, Havells India, Signify Innovations, Ledvance India, Surya Roshni, Crompton Greaves Consumer Electricals, Halonix Technologies, Central Power Research Institute, Bureau of Energy Efficiency, UL India, VOICE (consumer group)
Upcoming committee meetings	41st Sectional Committee Meeting (Virtual, 20 Aug 2026) — Completed; 42nd (Hybrid, 10 Dec 2026) — Upcoming; 43rd (Hybrid, 12 Feb 2027) — Upcoming

Use case: if a user asks 'who decides LED lamp standards', 'which companies are involved in setting this standard', or 'when is the next committee meeting', this data answers with real names and dates — a good trust signal for a technical/institutional query.

SECTION G — BIS Contacts (Electronics / CRS / Registration)

[CHUNK TYPE: structured-record] [RAG-TAG: contacts — SQL table, exact lookup by category]

Category	Contact Details
CRS Registration Department	Room 407, Manakalya Building, BIS, 9 Bahadur Shah Zafar Marg, New Delhi-110002 Phone: +91-11-23230856 Email: registration@bis.org.in
CRS Grievance (escalation)	DDG (MSCD & Registration) Email: ddgmscd@bis.gov.in
General BIS Helpdesk	9 Bahadur Shah Zafar Marg, New Delhi-110002 Phone: +91-76-6908-9323 Email: helpdesk@bis.gov.in
ETD 23 Committee (technical queries)	Member Secretary: Jainendra Kumar (via BIS Electrotechnical Department)
Original MeitY testing guidance issuer	Department of Electronics & IT, Electronics Niketan, 6 CGO Complex, Lodhi Road, New Delhi-110003

SECTION H — Official CRS FAQ (June 2018, Revision 12) — 53 Q&A

[CHUNK TYPE: qa-pairs] [RAG-TAG: faq, edge_cases, product_coverage] [Recommended: one chunk per Q&A pair for precise retrieval]

Source: Frequently Asked Questions, Updated (Revision 12): June 2018, published by MeitY/Electronics India. Covers Part I (Generic Issues, Q1–53) and Part II (Decisions Related to Product Coverage under the Order, Q1–46, overlapping numbering with Part I). Selected high-value entries most relevant to a consumer-facing/manufacturer-facing BIS assistant are extracted below; the full 53+46 question set should be ingested as individual chunks for complete coverage.

H.1 — Generic Registration & Testing Rules (selected)

Q: Manufacturing vs import date basis

A: The date of manufacturing applies for domestic goods; the date of landing of consignments in India applies for goods manufactured overseas, when checking against notified deadlines.

Q: Goods already in stock/market

A: Products manufactured or imported before the Order's effective date are not covered under the Order's ambit.

Q: Who gets the Unique Registration Number

A: The manufacturer is eligible to apply and receive the registration number, which is linked to manufacturer, factory location, product, and brand — not the importer.

Q: OEM products with existing CE/UL/FCC marks

A: Not accepted as-is. A valid test report (not older than 90 days) from a BIS-recognised lab, as per the notified Indian Standard, is mandatory — overseas/CB lab reports for other standards are not sufficient.

Q: Test report validity

A: Must not be older than 90 days as of the date of submission of both the online and physical application.

Q: Standard Mark placement

A: Must be placed on the product AND packaging; if not feasible on the product due to size, packaging-only is acceptable. E-labelling provisions exist for products with display screens.

Q: Multiple brands under one manufacturer

A: Each brand must be registered separately, though a common test report can be used if the product and manufacturing location are identical. All brands/model numbers must be listed in the test report.

Q: Registration per manufacturing location

A: A separate registration (and separate test report from that location) is required for each manufacturing location, even for the identical product.

Q: Who bears registration cost

A: The registered entity bears all costs, including those of surveillance.

Q: Retesting before renewal

A: Not specifically required — renewal is via application + fee at least one month before expiry. However, retesting may be needed if a test report per the latest upgraded standard is unavailable at renewal time (Order dated 13 Nov 2014, para 3).

Q: Supplier's test report for importers

A: Not acceptable — the importer/trading company cannot rely on a supplier's test report for self-declaration.

Q: Manufacturer's own test lab

A: Not entitled to BIS recognition — manufacturer in-house labs cannot be BIS-recognised test labs.

Q: Testing required per lot?

A: No — testing is not required for every production lot, though the registered entity remains responsible for ongoing compliance.

Q: Highly Specialized Equipment (HSE) exemption

A: Exempted if manufactured/imported under 100 units per model per year AND meets one of: 3-phase power; single-phase with current rating exceeding 16A; dimensions exceeding 1.5m x 0.8m; or weight exceeding 80kg.

Q: Penalty for unregistered sale

A: Liable for prosecution under Clauses 3(2) (deformation of non-conforming goods) and 4(3) (seizure) of the 2012 Order, plus 2(a)(2) of the amendment order dated 25 June 2013, and other BIS Act provisions.

Q: Safety critical components (indicative list)

A: Power cords, switches, safety isolating transformers, fuses, rechargeable batteries, PCBs, thermostats, X and Y capacitors, fusible resistors/varistors, plug/sockets/connectors, lamp holders, cells (for batteries), external/internal wire for LED Luminaires, insulation tape, electrolytic bulk capacitor. List is indicative, not exhaustive.

Q: Goods imported for demo/R&D/testing

A: Exempted from the Order per para 2(b) of the Gazette notification dated 05 July 2013.

H.2 — Product-Coverage Decisions Relevant to LED/Electronics (selected)

Q: Power banks — which standard applies?

A: Power banks with USB charging and NO internal electronic/charge-controlling circuit are treated as batteries and tested under IS 16046. Power banks WITH internal electronic circuitry continue to be treated as Power Banks under IS 13252 (Part 1), as notified in the CRS product list.

Q: Mobile phone chargers

A: Covered under the Order — chargers are notified products requiring their own registration.

Q: Consolidated products (e.g. laptop/mobile with battery+adaptor)

A: The overall product (laptop/mobile) bears its own registration number; the battery and adaptor, being independently notified, must ALSO carry their own independent registration numbers when separated from the main product.

Q: LED Bulbs — which cap types are covered

A: All cap types listed in Amendment No. 2 to IS 16102 (Part 1):2012 are covered under CRS — not limited to B22/E27/E14 only.

Q: Non-encapsulated LED Drivers (IS 15885 Part 2/Sec 13)

A: Not covered if not sold as a standalone product in the open market. Manufacturers relying on non-encapsulated drivers from suppliers are nonetheless encouraged to seek registration of those drivers.

Q: LED luminaire product-driver scenarios

A: A detailed decision table covers many driver/luminaire combinations (job-work sub-assembly boards = not covered; encapsulated drivers sold separately = covered; self-ballasted lamps with integral driver = no separate driver testing needed; etc.) — see full FAQ Part II Q40 table for exhaustive scenarios.

Q: Photometric test on LED luminaires/lamps

A: Exempted for CRS purposes generally, as the cross-referenced Indian Standard (IS 13383 Part 1) does not define parameters for LED-specific photometry; however, where photometric tests are part of the notified standard's existing scope (LED luminaires under IS 10322 Part 5, Sections 2–8 except 4, under CRO Phase III), they DO apply.

Q: Luminaire products not falling under any specific IS 10322 Part 5 section

A: Treated as covered under Section 1 (General Purpose Fixed Luminaire) by default. Baylight/Bulkhead fall under LED Flood Lights category. LED Portable lights are NOT covered under CRO.

Q: Batteries built into non-covered host products (e.g. shavers, trimmers)

A: Not covered under CRO if the host product itself is outside CRO scope and the battery is an inseparable part.

Q: Sealed Secondary Cells/Batteries as integral parts

A: Covered (mandatory BIS registration) with effect from 01.06.2016 even if integral to the host product, PROVIDED the host product is itself covered under the Order.

Note: this is a curated subset. The full FAQ document (53 Part-I + 46 Part-II questions) is available and should be chunked question-by-question for maximum RAG retrieval precision on specific edge-case queries not covered in this summary.

SECTION I — MeitY Applicant Testing Handbook (Product-Wise Test Submission Guide)

[CHUNK TYPE: process-template] [RAG-TAG: testing_submission, applicant_checklist] [SCOPE LIMITATION — READ CAREFULLY]

Source: 'Handbook of Information — Applicant Guidelines for submission of items for testing for participation in Compulsory Registration (CRS) Scheme notified by DeitY', issued by the Department of Electronics & IT, Ministry of Communication & IT, Government of India.

■ **IMPORTANT SCOPE LIMITATION:** This Handbook covers the applicant testing submission requirements **ONLY** for the **original 15 product categories** from the 2012 Order: Electronic Games, Laptop/Notebook/Tablet, Plasma/LCD/LED TVs (32"+), Optical Disc Players, Microwave Ovens, Visual Display Units, Printers/Plotters, Scanners, Wireless Keyboards, Telephone Answering Machines, Amplifiers, Electronic Musical Systems, Electronic Clocks, Set Top Box, and Automatic Data Processing Machines. **It does NOT include a LED Lamp-specific testing information sheet** — LED Lamps were added later via the 2014/2015 amendment and are not part of this document's product coverage. If the assistant is asked 'what information do I submit to the lab for LED lamp testing', this Handbook cannot directly answer that — use the general structural template below instead, or flag this as a documented gap.

I.1 — General Applicant Workflow (applies across all CRS products, including LED lamps)

1. Refer to the founding Order (Standing Order No. 8(14)/2006-IPHW, Gazette No. 1975, 03 Oct 2012) to confirm applicability to your product.
2. Refer to the official FAQ for further clarification (see Section H of this document).
3. Form series of products per the official Series Approval Guidelines, where applicable.
4. Keep the product-wise information sheet (see I.2 template below) ready for the lab.
5. Select a BIS-recognised testing laboratory that can test your specific product (see Section 4 of the base v1 document for the IS 16102 lab directory).
6. Contact the selected lab with your product details.
7. Discuss testing requirements with the lab and complete all paperwork (service request form, sample submission, technical documentation) and financial obligations.
8. Obtain a Probable Date of Completion (PDC) from the lab.
9. Cooperate with the lab during testing — provide clarifications and technical assistance (e.g. for mounting mechanical test items) as requested.
10. Collect the tested item, spare item, and test report on completion.
11. Retain the tested item in safe custody.
12. Initiate the BIS registration process (see Section E of this document).

I.2 — Reusable Applicant Information Template (structure identical across all CRS products)

Every product-wise sheet in this Handbook (and, by extension, the equivalent sheet a LED lamp applicant would need) follows this identical structure — useful as a generation template even though no LED-specific sheet exists in the source document:

- Applicant's name & address; Manufacturer's name & address (as in registration scheme)
- Test item description, Trade Mark, Model/Type reference
- Rated current (A) / Rated voltage (V)
- Overall size (W × H × D, mm) and mass (kg)
- Marked degree of protection to IEC 60529 (IPXX)
- Series Formation Basis (if applicable): models included, similarities, differences, worst-case model, max accessories used

- Product-specific supporting documents (varies — e.g. for a Laptop: CPU type, PCB layout, power adapter used, enclosure drawing, battery type; the equivalent LED-lamp-specific list would cover driver circuit, cap type, LED module details, and photobiological classification per IS 16108)
- List of Safety Critical Components table: for each component (SMPS/Adapter, enclosure insulating material, bridging resistor, capacitors, RF suppression X-Y capacitors, inductor, MOV/VDR, transformer, PCB material, plug/cable, connector, internal wire, LEDs/laser class, opto-coupler, EMI/EMC filter) — record Manufacturer/Trademark, Type/Model, Technical Data, Standard Mark(s) of Conformity, and Test Report reference

Standard notes appearing on every product sheet: (1) A qualified technical person must demonstrate the product and provide clarifications as requested. (2) The submitted sample must be a complete unit including adapter, additional mains adapter and enclosure if feasible, circuit schematic diagram, block diagram, PCB layouts, real-time clock battery details with protection, accessible ports, any special construction/additions, the Safety Critical Components list with test reports from a CBTL/BIS/IEC17025-approved lab, and the instruction & service manual.

Practical use for the assistant: when a user asks 'what do I need to submit for testing my [product]', retrieve the matching product sheet if it's one of the original 15 categories; for LED lamps (or any later-added category), use this I.2 template as a general-purpose answer while being transparent that a lamp-specific official sheet was not found in the ingested corpus.

SECTION J — Standard Scope Text: IS 16102 Part 1 & Part 2 (Edition Comparison)

[CHUNK TYPE: standard-clause-text] [RAG-TAG: scope_text, is_16102_part1, is_16102_part2, edition_diff] [HIGH VALUE — first real clause-level technical text, not just metadata]

Source: Clause 1 (Scope) and Clause 2 (References) of IS 16102 (Part 1):2026, IS 16102 (Part 1):2012 (Reaffirmed 2017), and IS 16102 (Part 2):2017 / :2026, as supplied directly by the user from the standard PDFs.

J.1 — Part 1 (Safety Requirements): 2012 vs 2026 Scope Comparison

Field	IS 16102 (Part 1):2012 (Reaffirmed 2017)	IS 16102 (Part 1):2026
Rated wattage	Up to 60 W	Up to 60 W (unchanged)
Rated voltage	D.C. supplies up to 250 V, or a.c. supplies up to 1000 V at 50 Hz	Above 50 V a.c. up to and including 250 V a.c. at 50 Hz, AND a d.c. voltage up to 1000 V — both a.c. and d.c. ranges now apply together, more precisely bounded
Cap types	Per Table 1 (referenced, not enumerated in scope clause)	Explicitly enumerated in scope: Edison screw caps E14, E27 and Bayonet caps B22d and B15d
Battery-incorporated lamps	Not explicitly addressed in 2012 scope text supplied	NEW — explicitly covers lamps incorporating rechargeable batteries (replaceable and non-replaceable); see Annex B
Additional-function lamps	Not addressed	NEW — explicitly extends to lamps with added functions: communication technologies, sensors, speakers, energy storage/battery charging
Photobiological safety	Not stated as included in scope note	NEW — explicitly stated as included (Note 2)
Batch/whole-product testing reference	Not specified in supplied text	References Annex D of IS 16103 (Part 1) for whole-product/batch testing recommendations

Part 1 (2026) — Full clause-1 scope text (paraphrased, structure preserved): Specifies safety and interchangeability requirements, plus test methods/conditions, for self-ballasted LED lamps for domestic/general lighting, rated up to 60 W, above 50 V a.c. up to 250 V a.c. at 50 Hz and up to 1000 V d.c., with E14/E27 Edison screw caps and B22d/B15d bayonet caps. Batch/whole-product testing follows Annex D of IS 16103 (Part 1). Scope explicitly includes photobiological safety, battery-incorporated lamps (Annex B), and lamps with added communication/sensor/speaker/energy-storage functions.

Part 1 (2026) — Full References list (Clause 2, Annex A): IS 8913:2025/IEC 60360:1998 (lamp cap temperature rise measurement); IS 9206:1979 (tungsten filament lamp cap dimensions); IS 10322 (Part 1):2014 (Luminaires — general requirements and tests); IS 1401:2008/IEC 61032:1997 (protection by enclosures — verification probes); IS 15518 (Part 1):2004 (incandescent lamp safety); IS 15885 (Part 1):2011 (lamp controlgear safety); IS 16108:2012/IEC 62471:2006 (photobiological safety of lamps); IS 16661:2019/IEC/TR 62778:2014 (blue light hazard assessment); IS/IEC 60529:2001 (IP code); IS/IEC 60695-2-10/-11/-12/-13:2021 (glow-wire fire hazard test series); ISO 4046-4:2016 (paper/board vocabulary); IEC 60061-3 (lamp cap/holder gauges); IEC 61347-1:2015/AMD1:2017 (lamp controlgear general/safety requirements).

J.2 — Part 2 (Performance Requirements): 2017 vs 2026 Scope Comparison

Field	IS 16102 (Part 2):2017	IS 16102 (Part 2):2026
Rated power	1 W to 25 W	Up to 60 W — significantly widened range, now aligned with Part 1 safety scope
Rated voltage	Above 50 V a.c. up to and including 250 V a.c. at 50 Hz only	Above 50 V a.c. up to 250 V a.c. at 50 Hz, AND d.c. voltage up to 1000 V — d.c. now explicitly covered
Integral means for operation	"integral means for stable operation"	"integral ballast meant for stable operation" — terminology shift to "ballast"
Battery-incorporated lamps	Not addressed	NEW — explicitly covers lamps with rechargeable batteries (replaceable/non-replaceable)
Additional-function lamps	Not addressed	NEW — explicitly extends to communication/sensor/speaker/energy-storage-equipped lamps
Rated life / lumen maintenance approach	No explicit maximum rated life figure; uses lumen-maintenance codes at a defined finite test time instead of life validation; a general extrapolation method was "under consideration"	NEW — specifies test requirements for a maximum DECLARED rated life of up to 50,000 h; verification beyond 50,000 h still not confidently possible (extrapolation method still unavailable)
Operating temperature range	Ambient air –10°C to 40°C (at 90–110% rated voltage)	WIDENED — ambient air –10°C to 50°C (at 90–110% rated voltage) — 10°C higher upper bound
Claimed different operating conditions	Lamps must be tested and start/operate satisfactorily under claimed different conditions	More explicit 3-part requirement: (a) must be tested under claimed conditions; (b) must start/operate satisfactorily; (c) must meet performance claims under those conditions, which may differ from general measurement conditions in Annex B-1
Colour scope	White light only (inorganic LEDs); excludes tinted/coloured light and OLEDs	Unchanged — same exclusions (confirmed by continuity in supplied text)

Practical significance for the assistant: the 2026 revision of Part 2 substantially widens the wattage range (25W → 60W) to match Part 1, formally caps declared life at 50,000 hours, raises the upper operating-temperature bound by 10°C, and — like Part 1 — extends coverage to battery-incorporated and multi-function (smart/connected) LED lamps for the first time. A user asking 'is my 40W smart LED bulb with Bluetooth speaker covered' would have been OUT of scope under the 2017 edition (25W cap, no multi-function provision) but IS covered under the 2026 edition on both counts.

Part 2 (2017) — Full References list (for historical/legacy queries): IS 1885 (Part 16/Sec 1):1968 (lighting vocabulary); IS 2418 (Part 2) (tubular fluorescent lamp performance, under print); IS 6873 (Part 5):2012 (radio disturbance limits — lighting equipment); IS 9000 (Part 14/Sec 1) (thermal shock test procedure); IS 14700 (Part 3/Sec 2):2008 (harmonic current emission limits); IS 14700 (Part 4/Sec 7):2006 (harmonic measurement technique guide); IS 16101:2012 (LED/LED module terms & definitions); IS 16102 (Part 1):2012 (the paired safety standard); IS 16105:2012 (lumen maintenance measurement method); IS 16106:2012 (electrical/photometric measurement of SSL products); IS/IEC TR 61341:2010 (centre beam intensity/beam angle measurement for reflector lamps).

Not yet supplied: the full body clauses beyond Scope/References (e.g. Part 1 Clauses 3 onward — construction, marking, dielectric/mechanical/thermal test requirements; Part 2 Clauses 3 onward — luminous flux, efficacy, colour rendering, lumen maintenance test procedures, pass/fail thresholds) are still outside this dataset. These are the deepest technical-question layer and would need the remaining clause text if the assistant is asked for specific numeric test limits.

SECTION K — Updated Assessment: Is This Enough?

Yes — this dataset is now comprehensive for a strong working prototype covering the LED Lamp / IS 16102 product category plus generalizable CRS process knowledge. You now have:

- ✓ Standard identity, classification, and supersession chain (both parts, both old and new editions)
- ✓ The actual legal Order text establishing CRS and its core prohibition/enforcement clauses (2012, bilingual)
- ✓ The exact 2026 concurrent-running/withdrawal dates across 13 related lighting standards — real freshness-check data
- ✓ Standard Mark specification and labelling rules, including R-number format and exact monogram/colour spec
- ✓ Full registration, renewal, and inclusion process with real timelines
- ✓ The technical committee (who governs the standard) plus upcoming meeting dates
- ✓ Real testing lab directory with pricing (58 labs total across both parts)
- ✓ Real registered-manufacturer sample records with R-numbers and status
- ✓ Direct BIS contact details, segmented by query type
- ✓ 53+46 official FAQ entries covering dozens of product-classification edge cases (power banks, chargers, batteries, LED luminaires vs modules vs lamps)
- ✓ The official applicant testing submission template/workflow, transferable across product categories

The one remaining gap is the actual clause-by-clause technical text of IS 16102 itself (specific safety test parameters, specific performance thresholds like lumen maintenance %) — this requires the full standard PDF from the BSB Edge sales portal (pending your account activation). This is only needed if your assistant must answer deep technical/engineering questions; for certification-guidance, scheme-explanation, freshness-checking, product-classification edge cases, and lab-referral use cases (which cover the large majority of realistic user queries for this problem statement), this dataset is complete and demo-ready.